The missing linear component of the cubic singlet
Exact charge-sector channels and their double-scaling weight

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Abstract

We calculate the first Gaussian coupling chaos of the raw Hermitian cubic derivative singlet. It is not generally a conserved function of the Hamiltonian. Its exact finite-size map is diagonal in standard $U(N)$ irreducible tensor channels, with coefficients given below by binomial expressions. After subtracting the ensemble projection onto the Hamiltonian in each charge sector, the residual has double-scaled norm $64\mathcal{J}_s^6 e^{-8\lambda}$. This identifies a surviving component omitted by the third-chaos projection and sets the next spectral calculation.

1 What is added to the cubic construction

Use standard CAR, $H = H_s$, $L = [H, \cdot]$, and

$$A = \sum_i (Lc_i)^\dagger Lc_i, \quad T_3 = \tfrac{1}{2}\left(\sum_i c_i^\dagger L^3 c_i + \text{h.c.}\right).$$

The exact identity $M_3 = T_3 + \frac{3}{2}LA$ and the previously constructed q^3 channel R give

$$\boxed{M_3 = R + aS + bZ + \bar{t}(Q_f)H + F + \tfrac{3}{2}LA,} \tag{1}$$

where $S = \mathcal{W}_3\{H, A\}/2$, $Z = \mathcal{W}_3 H^3$, $R = \mathcal{W}_3 T_3 - aS - bZ$, and

$$a = -\frac{(1-q)^2}{1+q^2}, \qquad b = \frac{2\lambda(1-q)(1+q^3)}{(1+q^2)(1+q+q^2)}, \quad q = e^{-4\lambda}.$$

The new terms are defined exactly at finite N:

$$\mathcal{W}_1 T_3 = \bar{t}(Q_f)H + F, \qquad \mathbb{E}_J \operatorname{tr}_m(HF) = 0. \tag{2}$$

Here tr_m is the normalized trace on charge m; $\bar{t}(m)$ is a deterministic projection coefficient. It follows that $\bar{t}(Q_f)H$ is conserved in each realization. F can still contain static components; (2) is not a projection onto the entire commutant.

In the full $\beta = \mu_{\text{chem}} = 0$ annealed trace, first take $N, p \to \infty$ at $\lambda = p^2/N > 0$, with $\langle H^2 \rangle \to \sigma^2 = \mathcal{J}_s^2/\lambda$. Our norm result is

$$\boxed{\lim_{\text{DS}} \langle F^2 \rangle = 64\mathcal{J}_s^6 q^2 > 0.} \tag{3}$$

The earlier R has $\langle R^2 \rangle \sim 32\mathcal{J}_s^6/\lambda$ as $\lambda \to 0^+$, so $\langle F^2 \rangle / \langle R^2 \rangle \sim 2\lambda$. The raw derivative family therefore has an additional component at the first relative order in this limit.

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2 Exact coefficients from covariance channels

Write the coupling covariance in standard CAR units as $\mathbb{E}J_{IJ}J_{KL} = w\delta_{IL}\delta_{JK}$. The normal-ordered p-body Hamiltonian decomposes into the established, multiplicity-free $U(N)$ tensor sectors

$$H_m = \sum_{\nu=0}^{\min(p,m,N-m)} H_m^{(\nu)}.$$

Define the following quantities; every binomial is zero outside its allowed range:

$$C_m = w\binom{m}{p}\binom{N-m+p}{p}, \tag{4}$$

$$\kappa_\nu(m) = w\binom{m-\nu}{p}\binom{N-m+p-\nu}{p}, \tag{5}$$

$$j_c(m) = w\binom{m-1}{p}\binom{N-m+p}{p}, \tag{6}$$

$$j_-(m,\nu) = w\binom{m-\nu}{p}\binom{N-m+p-\nu+1}{p}, \tag{7}$$

$$j_+(m,\nu) = w\binom{m-\nu-1}{p}\binom{N-m+p-\nu}{p}, \tag{8}$$

$$\ell_\nu = \frac{\nu(N-\nu-p+1)}{N-2\nu+1}. \tag{9}$$

$C_m = \mathbb{E}K_m^2$ is scalar, and $\mathbb{E}K_m H_m^{(\nu)} K_m = \kappa_\nu(m) H_m^{(\nu)}$ for an independent copy K of H. These covariance eigenvalues are standard embedded-ensemble results [2, 1]. The $j_\pm$ are the corresponding particle-removal channels.

The first Gaussian chaos is

$$(\mathcal{W}_1 T_3)_m = \sum_\nu t_\nu(m) H_m^{(\nu)}, \tag{10}$$

with the exact coefficient

$$\begin{aligned} t_\nu(m) = &-(m+2p)C_{m-1} + (m-3p)C_m + 3p\, j_c(m) \\ &+ (m-p)\kappa_\nu(m-1) - m\kappa_\nu(m) \\ &+ 3\{\ell_\nu j_-(m,\nu) + (p-\ell_\nu) j_+(m,\nu)\}. \end{aligned} \tag{11}$$

This result keeps the finite-N rank dependence. Replacing all κ_ν/C_m by the common leading crossing factor too early would erase the residual (3).

For example, at $(N,p,m) = (5,2,3)$ and $w = 1$,

$$(t_0, t_1, t_2) = (-96, -167/2, -88).$$

Distinct coefficients on noncentral sectors allow nonconservation. A deterministic Gaussian-integer Hamiltonian specified with the saved validation data gives the exact witness

$$\mathrm{tr}_3[H, \mathcal{W}_1 T_3]^\dagger [H, \mathcal{W}_1 T_3] = \frac{8933634}{5} > 0. \tag{12}$$

Since the subtraction in (2) commutes with H, the same witness applies to $[H, F]$. It tests a finite system, not a double-scaling dynamical limit.

3 Derivation through particle-removal tensors

For a cubic polynomial P, $\mathcal{W}_1 P(H)$ is the sum of its three terms containing two copies of an independent Gaussian K and one H, averaged over K. Apply this to

$$L^3 c = H^3 c - 3H^2 cH + 3HcH^2 - cH^3.$$

Let $\mathcal{J}_m(Y) = \mathbb{E} K_{m-1} Y K_m$ and $\Phi(Y) = \sum_i c_i^\dagger Y c_i$. Homogeneity gives $\Phi(H_{m-1}) = (m-p)H_m$ and $\mathcal{J}_m(c_i) = j_c(m) c_i$. Grouping the three contractions in each monomial gives

$$\begin{aligned}(\mathcal{W}_1 M_3)_m = {} & [-(m+2p)C_{m-1} + (m-3p)C_m + 3pj_c]H_m \\ & + \Phi(\mathbb{E} K_{m-1} H_{m-1} K_{m-1}) - m\mathbb{E} K_m H_m K_m \\ & + 3\sum_i c_i^\dagger \mathcal{J}_m([c_i, H]).\end{aligned} \tag{13}$$

For a traceless charged tensor with r creations and s annihilations, input charge m, the covariance multiplier is

$$\mathcal{J}(Y) = w \binom{m-s}{p} \binom{N-m+p-r}{p} Y. \tag{14}$$

It follows by normal ordering a highest-weight disjoint-index tensor: the paired K indices have $m-s$ available occupied slots and $N-m+p-r$ available complementary slots. Equivariance extends the result to its irreducible tensor space.

To resolve $[c_i, H^{(\nu)}]$, write $H_m^{(\nu)} = f_m O_\nu$ with $f_m = \binom{m-\nu}{p-\nu}$ and O_ν a primitive traceless ν-body tensor. Set $u_i = c_i O_\nu$, $v_i = O_\nu c_i$. The difference $u_i - v_i$ belongs to the lower charged sector $(r,s) = (\nu-1, \nu)$. Only this and the upper sector $(\nu, \nu+1)$ occur; multiplicity freedom makes the lower projection proportional to $u - v$. In unnormalized sector trace, dividing all inner products by $\mathrm{Tr}_m\, O_\nu^\dagger O_\nu$,

$$\|u\|^2 = m, \quad (u,v) = m - \nu, \quad \|v\|^2 = \frac{(N-m+1)(m-\nu)}{N-m-\nu+1}.$$

Thus the lower projection of u is $(N-m-\nu+1)(u-v)/(N-2\nu+1)$. Since $f_{m-1}/f_m = (m-p)/(m-\nu)$, contraction with $\sum_i c_i^\dagger$ gives weights ℓ_ν and $p-\ell_\nu$ in the lower and upper charged sectors. Using (14) in (13) proves (11). For $\nu = 0$ the lower weight is zero, understood without a negative-rank tensor.

All coefficients are real. The resulting map preserves Hermiticity, so $\mathcal{W}_1 M_3 = \mathcal{W}_1 T_3$ and $\mathcal{W}_1 LA = 0$. The displayed derivation therefore also checks that the exact time-descendant subtraction does not hide another first-chaos term.

4 Rank weights and the surviving norm

Let $d_\nu = \binom{N}{\nu}^2 - \binom{N}{\nu-1}^2$, and use the standard covariance eigenvalue notation [1]

$$\Lambda^\nu(N,m,r) = \binom{m-\nu}{r}\binom{N-m+r-\nu}{r}.$$

The normalized Hamiltonian weight of channel ν is

$$\omega_\nu = \frac{d_\nu \Lambda^\nu(N,m,m-p)}{\binom{N}{m}\Lambda^0(N,m,p)}, \qquad \sum_\nu \omega_\nu = 1. \tag{15}$$

Different tensor components are independent Gaussian variables and orthogonal in the ensemble Hilbert–Schmidt inner product. Consequently

$$\bar{t}(m) = \sum_\nu \omega_\nu t_\nu(m), \tag{16}$$

$$\mathbb{E}\,\mathrm{tr}_m\, F^2 = C_m^3\ \mathrm{Var}_\omega\left(\frac{t_\nu(m)}{C_m}\right). \tag{17}$$

At half filling $m = N/2$, put $s = p - \nu$. The exact adjacent-weight ratio is

$$\frac{\omega_{\nu-1}}{\omega_\nu} = \left(\frac{\nu}{N-\nu+1}\right)^2 \frac{1-\left(\frac{\nu-1}{N-\nu+2}\right)^2}{1-\left(\frac{\nu}{N-\nu+1}\right)^2} \frac{m-\nu+1}{p-\nu+1}\frac{N-p-\nu+1}{N-m-\nu+1}. \tag{18}$$

For fixed s this tends to $\lambda/(s+1)$. For sufficiently large N and all $0 \le s < p$, it is bounded by $32\lambda/(s+1)$. Normalization and this factorial tail give

$$s \implies \mathrm{Poisson}(\lambda) \tag{19}$$

with convergence of every polynomial moment.

Writing $c_\nu = t_\nu/C_m$, direct expansion of (11) gives

$$c_{p-s} - c_p \longrightarrow 8\lambda q\, s. \tag{20}$$

One way to see the coefficient is to write $c_\nu = B_0 + (\kappa_\nu/C_m)E_\nu$, where B_0 is independent of ν. At half filling $E_p/p \to 2$, $E_{p-s} - E_p = O_s(p/N)$, $\kappa_p/C_m \to q$, and $\kappa_{p-s}/\kappa_p = 1 + 4\lambda s/p + O_s(p^{-2})$. The rational factors in (11) also bound $|c_{p-s} - c_p| \le C_\lambda(1+s)$ uniformly for $s \le p$. Together with the factorial tail, this justifies taking the variance, rather than only a pointwise coefficient limit:

$$\mathrm{Var}_\omega(c_\nu) \longrightarrow 64\lambda^3 q^2. \tag{21}$$

5 Full trace, scale comparison, and numerical checks

The half-filled result extends to the full $2^{-N}\,\mathrm{Tr}$ state. For m/N in a compact neighborhood of $1/2$, the ratio and slope estimates above are uniform, with continuous density dependence. The binomial charge distribution concentrates at $1/2$. Also C_m is maximal near $m = (N+p)/2$ and $C_{\max}/C_{N/2} = \exp[O(p^3/N^2)] \to 1$. Hence $C_m/\sigma^2 \to 1$ on the concentrating charge window. Outside a fixed density window the binomial charge tail is exponentially small, while (11) bounds the coefficients by $O(NC_{\max})$; these tails do not contribute to (17). Equations (17) and (21) then give (3).

The cubic-chaos result from the preceding stages is

$$\langle R^2\rangle_{\mathrm{DS}} = \sigma^6 V, \qquad V = \frac{4\lambda(1+q)[1-q^3+4\lambda q^2(2q^2-q+2)]}{(1+q^2)(1+q+q^2)}.$$

Therefore the exact double-scaled equal-time ratio is

$$\frac{\langle F^2\rangle_{\mathrm{DS}}}{\langle R^2\rangle_{\mathrm{DS}}} = \frac{64\lambda^3 q^2}{V} = 2\lambda + O(\lambda^2). \tag{22}$$

Gaussian chaos makes the linear and cubic components orthogonal at equal time. Real-time evolution depends on the same random H and does not preserve that chaos decomposition; their time-dependent cross correlators must be computed.

The channel formula was checked against direct Gaussian polarization in occupation-basis CAR matrices for $(N,p) = (3,1),(4,2),(5,2),(6,2),(6,3)$: 42 allowed sector/rank coefficients agree exactly with bounded Gaussian-integer arithmetic. Independent tests cover 31 charged covariance eigenvalues, 199 exact rational weight normalizations and ratios, three symbolic projection identities, and the nonzero witness (12).

The full rank sum, with no rank cutoff, was evaluated at 80 and 110 decimal digits for three λ values and $p = 16, 32, 64, 128$. The table shows the relative difference of the variance from (21):

λ	$p=16$	$p=32$	$p=64$	$p=128$
0.10	-13.97%	-6.94%	-3.45%	-1.72%
0.25	-7.76%	-3.51%	-1.66%	-0.81%
0.50	$+3.54\%$	$+2.50\%$	$+1.41\%$	$+0.74\%$

The finite-p differences are retained rather than treated as numerical error. Precision comparisons are below 10^{-55} in relative variance. The limiting statement is supported by the ratio and uniform-bound argument, not inferred from this table.

6 Physical interpretation and the next calculation

The original derivative operator is now decomposed into explicit pieces in (1). The known q^3 channel remains intact. The missing linear part has an exactly known finite-N map; its energy-orthogonal remainder survives double scaling in norm, and finite-size examples demonstrate nonconservation.

The next question is its *dynamical* spectral measure. A positive equal-time residual alone does not establish a new propagating particle or rule out additional conserved projections. In particular, identifying a q-probe kernel for F requires same-coupling mixed moments or an explicit covariance reduction. It cannot be concluded merely because $\nu/p \to 1$. The rank-dependent terms in (11) show exactly which subleading coefficients must be kept in that calculation.

The concrete next step is to evaluate $\mathbb{E}\,\mathrm{tr}(FL^2F)$ and the two-energy strength density using the resolved covariance channels, including H–F cross contractions. This will determine how the residual fits into the full raw-M_3 correlator. We can then compute mixed correlators with the already established $d = 2$ and $d = 3$ singlets, to supply new interaction data for the Susskind program. Their physical mass assignment still needs the spatial dictionary.

Source boundary. The irreducible decomposition, covariance eigenvalues, dimensions and channel weights used here come from embedded-ensemble representation theory [1, 2]. The charged multiplier was obtained by the same normal-order counting and explicitly checked. The new calculation recorded in this note is the application to $\mathcal{W}_1T_3$, its finite-sector coefficients, and the energy-orthogonal residual limit. The source papers' spectral-density correlation functions are not confused with the transition-strength correlator of F. No priority claim is made for standard representation theory.

All scripts and raw outputs are local. Earlier cubic-kernel and other scientific artifacts were preserved byte-for-byte. The six-page PDF was actually compiled and visually checked. No Git commit, push or archive was made; the complete de Sitter research objective remains active.

References

[1] V. K. B. Kota, *Bivariate moments of the two-point correlation function for embedded Gaussian unitary ensemble with k-body interactions*, arXiv:2208.11312v2, Sec. IV, Eqs. (28)–(39).

[2] J. Tall and S. Tomsovic, *Double-scaled bosonic and fermionic embedded ensembles, complex SYK, and the dual Hilbert space*, arXiv:2604.14522v2, Sec. 2.1, Eqs. (2.8)–(2.19).